ARP HANDLING IN SDN NETWORKS

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OBJECTIVE :-

To implement ARP request and reply handling using an SDN controller and enable communication between hosts.

TOOLS USED :-

- Mininet
- Ryu Controller
- Ubuntu

METHODOLOGY :-

- Created a Mininet topology with 2 hosts and 1 switch
- Connected switch to Ryu controller
- Implemented ARP packet interception in controller
- Used MAC learning for forwarding
- Installed flow rules dynamically

WORKING :-

When h1 sends ping to h2:

- ARP request is generated
- Controller intercepts ARP packet
- Logs source and destination IP
- Switch forwards packet
- Communication established

RESULT :-

- ARP packets successfully intercepted
- Communication between hosts established

- Flow rules installed dynamically

1) python3 -m venv ryu-env
   source ryu-env/bin/activate
   ryu-manager arp_controller.py

```
(ryu-env) vboxuser@Ubuntu:~$ source ryu-env/bin/activate
ryu-manager arp_controller.py
loading app arp_controller.py
loading app ryu.controller.ofp_handler
instantiating app arp_controller.py of ARPController
instantiating app ryu.controller.ofp_handler of OFPHandler
```

```
(ryu-env) vboxuser@Ubuntu:~$ source ryu-env/bin/activate
ryu-manager arp_controller.py
loading app arp_controller.py
loading app ryu.controller.ofp_handler
instantiating app arp_controller.py of ARPController
instantiating app ryu.controller.ofp_handler of OFPHandler
ARP Packet: 10.0.0.1 -> 10.0.0.2
ARP Packet: 10.0.0.2 -> 10.0.0.1
```

2) sudo mn --topo single,2 --controller=remote,port=6653 --switch ovsk,protocols=OpenFlow13
   h1 ping -c 4 h2

```
mininet> h1 ping -c 4 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.118 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.183 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.239 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.093 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3110ms
rtt min/avg/max/mdev = 0.093/0.158/0.239/0.057 ms
mininet>
```

3) net

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo:  s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet>
```

4) h2 ifconfig h2-eth0 down → disabling host

h1 ping -c 3 h2

```
mininet> h2 ifconfig h2-eth0 down
mininet> h1 ping -c 3 h2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.

--- 10.0.0.2 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2061ms

mininet>
```

0 packets received, 100% packet loss. Host is disabled.